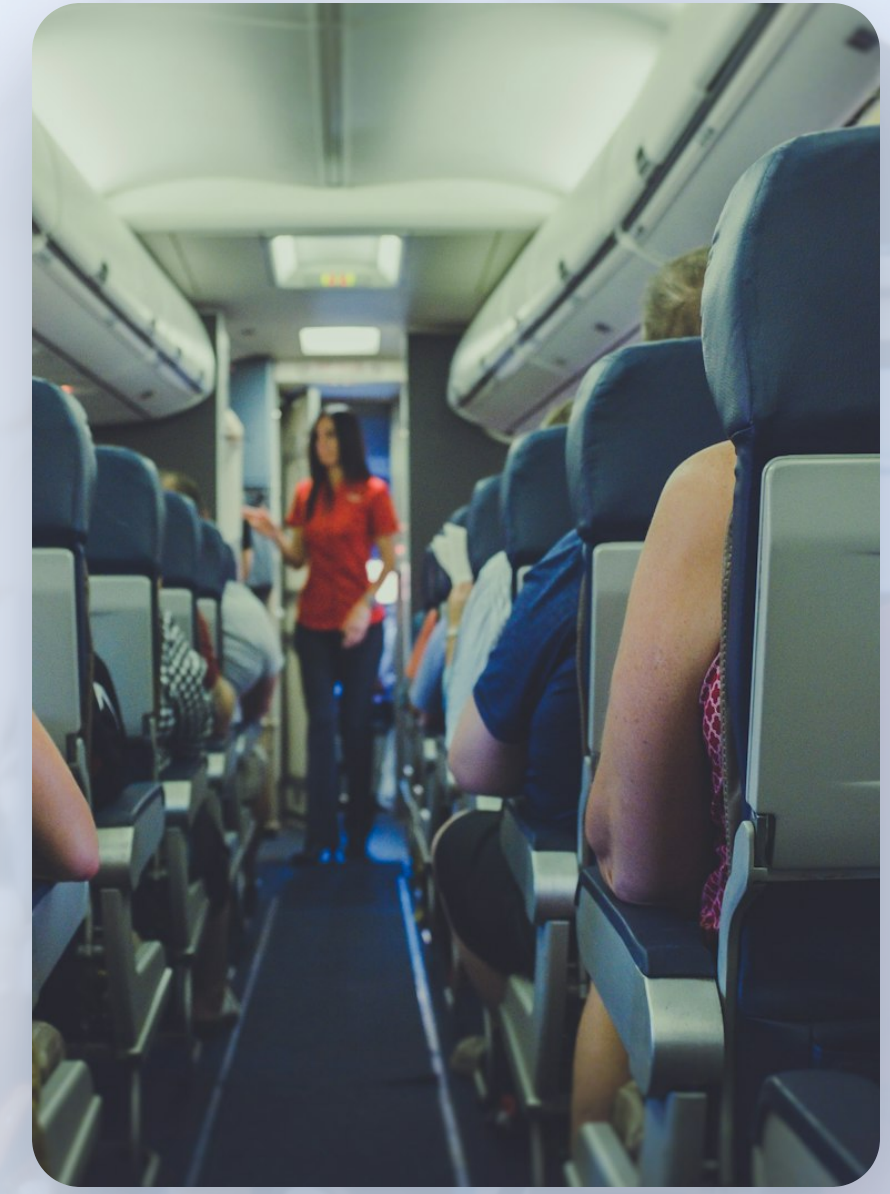
HearMe

Because every passenger deserves to be heard.

Helping passengers and cabin crew understand each other.

In any language. In a noisy cabin.
Even with no internet.



Submitted by Nidhish Bansal · Dhruv Mittal · Dibyanshu Samal

Idea submission · Airline passenger–crew communication



Sharda, 68 · speaks only Hindi

She is flying from Delhi to Tokyo to see her son. She is a little hard of hearing and needs warm water for her tablets on time.

Sharda is an example persona.

THE PROBLEM

Two people, two accents, one loud cabin.



Signs and announcements are only in **Japanese and English**.



The call button tells the crew that **someone** needs help. Not **what** they need.

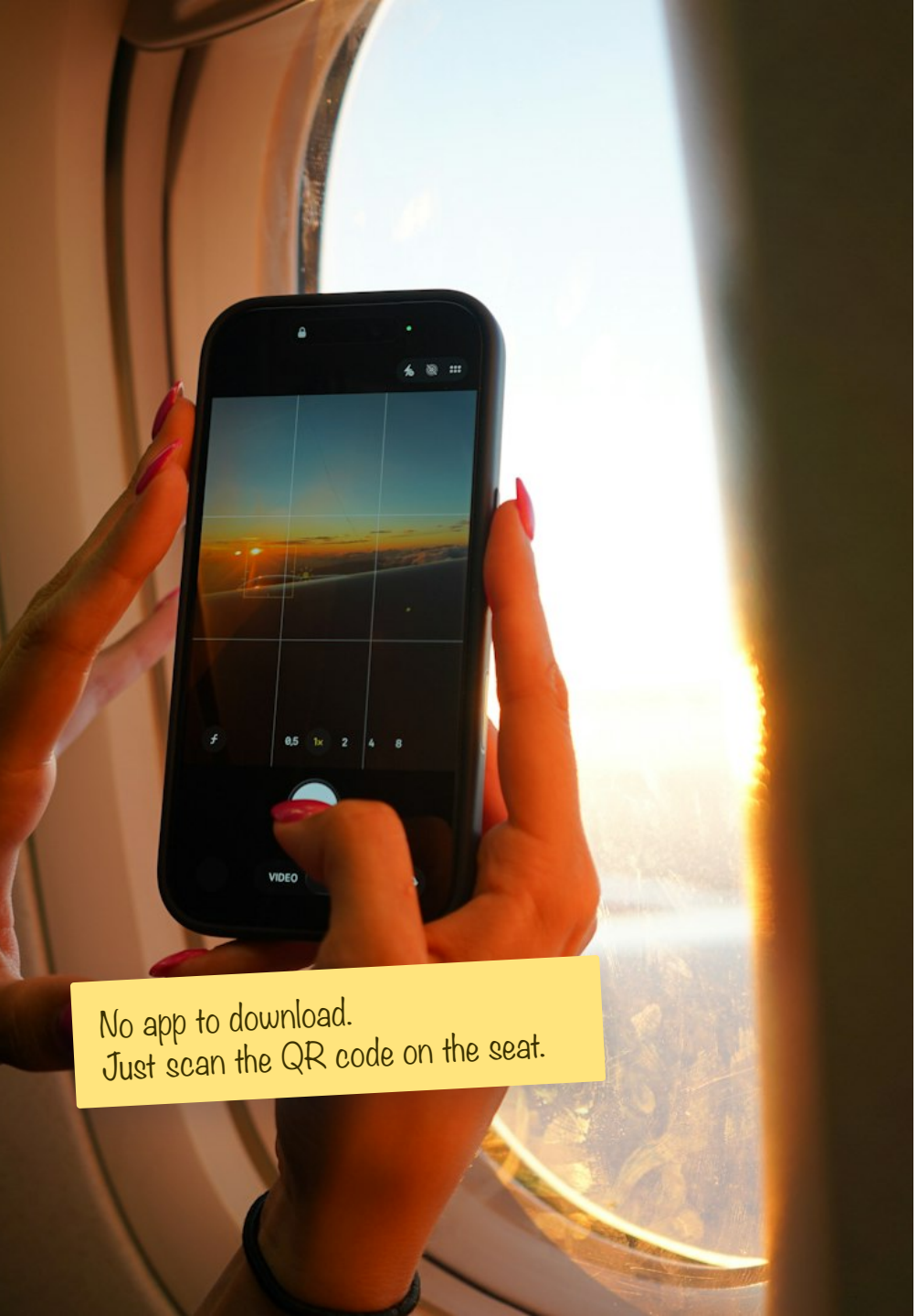
English is a second language for Sharda **and** for the crew member. After one “pardon?” too many, she simply stops asking.

5.1 million
visitors to Japan from India and Southeast Asia in 2025

75–85 dB
of cabin noise in flight, as loud as a busy road

Over 1 in 4
people over 60 have disabling hearing loss

Sources: JNTO visitor arrivals 2025 (preliminary) · WHO hearing loss fact sheet · in-flight cabin noise studies



OUR IDEA

Tap it, say it, or point at it.

1. Sharda taps a picture or just speaks Hindi. No phone? She points at the printed card in the seat pocket.
2. The crew tablet shows **“23C · hot water for medicine”** in Japanese, before anyone walks over.
3. The answer comes back in Hindi: on screen, as a picture, and as a voice in her earphones.

The trick

We don’t translate every word. Most requests on a plane are the same 64 things, like water, a blanket, the toilet or medicine. Matching to one of those still works with strong accents and engine noise. Not sure? The app asks with pictures.

Check-in
Language and needs saved once

Boarding
Safety briefing in pictures

In the air
Requests and announcement captions

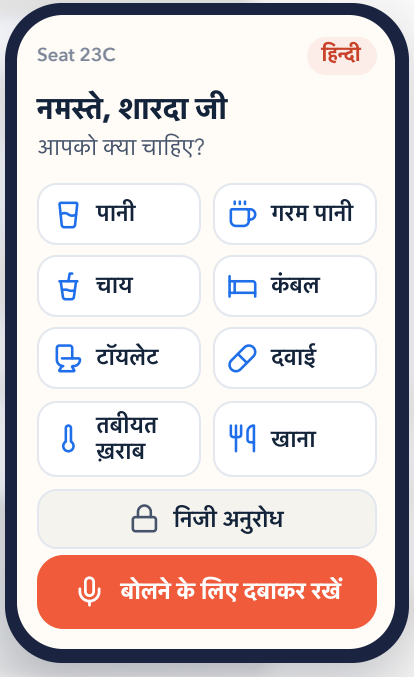
Arrival
Medical and wheelchair handover

WHAT IT LOOKS LIKE

One request, start to finish

Concept mockups · design only, not built yet

Sharda needs warm water for her tablets. Here is what she and the crew see.



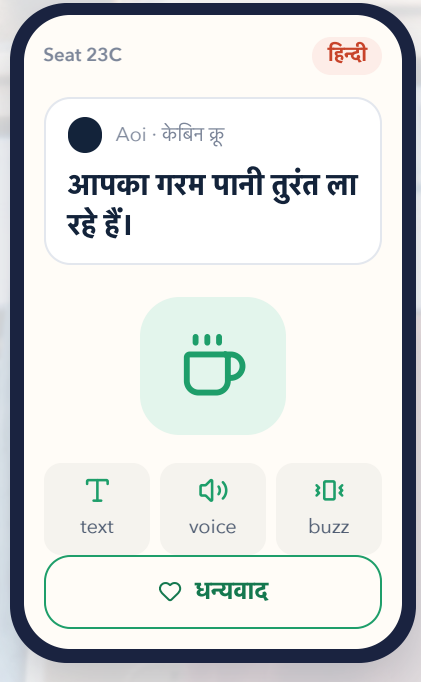
1. Opens in Hindi with big picture buttons.



2. Noise cut her words, so it asks with pictures.



3. The crew know what she needs before walking over.



4. The answer in Hindi, also read out in her earphones.

HOW WE'LL BUILD IT

Everything runs inside the plane

The AI runs on the crew's tablet, so it works with no internet at all.

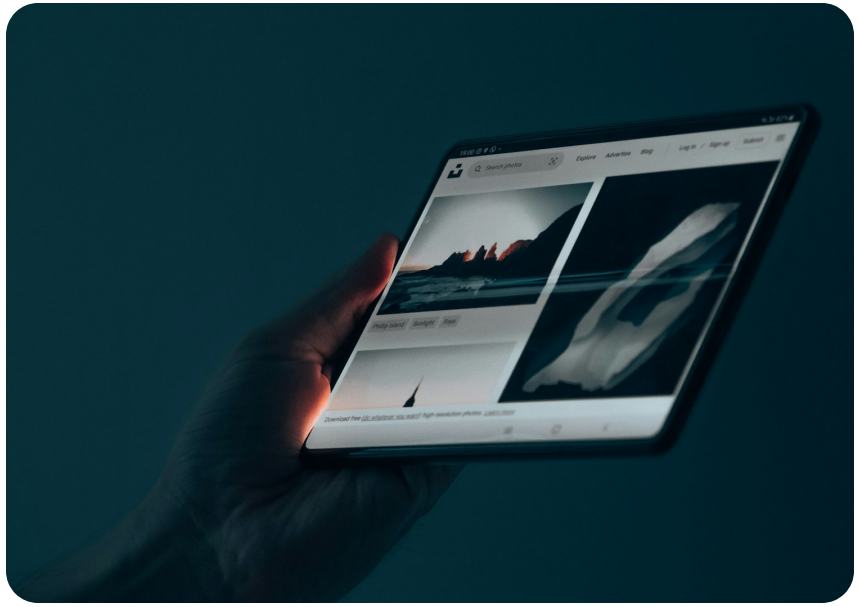
Passenger's phone
A web app that opens from the seat QR code

→

Crew tablet
Runs the speech AI and keeps the request list, fully offline

→

Crew app
Requests by seat, with one-tap replies



What happens when Sharda speaks

1. **Remove engine noise**
   DeepFilterNet
2. **Turn speech into text**
   Whisper (runs offline)
3. **Match it to 1 of 64 needs**
   multilingual-e5
4. **Not sure? Ask with pictures**
   if confidence < 85%

Built with

React web app (PWA) · Python + FastAPI · WebSockets · Whisper · multilingual-e5 · SQLite

Works with no internet
Phones talk to the crew tablet over the cabin Wi-Fi only

Answers in under 2 seconds
Our target on a normal tablet

Voice is never saved
Only "seat 23C asked for water" is kept

Safety messages are checked by people
No machine translation for safety



No passenger should fly unheard.

HearMe · in any language, from any seat

WHY IT CAN WORK

Ready for real flights

- **No new hardware**
  Passengers' own phones and the crew's existing tablets
- **No internet needed**
  A printed picture card works even if the Wi-Fi is down
- **Uses data airlines already have**
  Language and needs travel with the booking
- **Easy for crew**
  Most replies take two taps, in Japanese

Our plan

NOW · BUILT
Working prototype
Offline AI on one laptop: 94% of typed and 83% of spoken requests understood in cabin noise

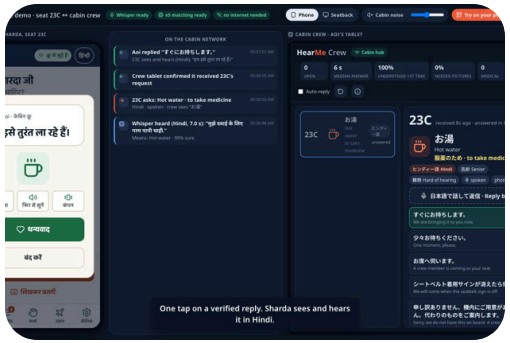
NEXT
3-month pilot
Tokyo–Delhi and Tokyo–Bangkok flights

THEN
All routes
India and Southeast Asia first, 14 languages

What we'll measure

Faster help · Fewer repeats · Safety briefing understood · Happier passengers

Easy to grow: adding a new language means translating about 5,000 words. No new code.



Photos from Unsplash by Quilia, Tanmay Abhay Mahajan, Ivy Lu, Tymur Kuchumov, Yosuke Ota, Suhyeon Choi, Egor Komarov, Natali Quijano and Kazuo Ota.

WHAT IT COSTS

No new hardware. One subscription per aircraft.

THE AIRLINE NEEDS

Almost nothing new

- Passengers' own phones
- The crew's existing tablet, or a small cabin PC
- Printed picture cards in seat pockets
- No internet, no cloud bill, no per-request AI fees

WE CHARGE (ASSUMPTION)

~$150 per aircraft / month

- Software, updates and new languages
- Crew dashboard and flight reports
- A 100-aircraft airline: ~$180k a year
- A new language: a data file plus native review, no code

THE AIRLINE GETS

Measurable service

- Answer time and "understood first time" per flight
- Faster medical response, with a handover sheet
- Accessibility for passengers who are hard of hearing or don't speak the language
- A pilot proves the numbers on real routes

Prices are our working assumptions for a pilot conversation, not quotes. The value side will be measured in a 3-month pilot with airline data.